📘 AI/ML Internship – Day 66 Notes & Tasks

🟢 Module 8: Retrieval-Augmented Generation (RAG) – Building Chatbots with Custom Knowledge

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📌 Part 1: Recap of Day 65

Yesterday we learned:

✅ Conversation Memory

✅ Chat History

✅ Context-Aware Responses

✅ AI Chatbot with Memory

Today we will learn:

- What is RAG?
- Why RAG is needed
- How RAG works
- Difference between ChatGPT and RAG
- Real-world applications

📌 Part 2: What is RAG?

Definition

Retrieval-Augmented Generation (RAG) is an AI technique that combines **information retrieval** with a **Large Language Model (LLM)** to generate accurate and context-aware answers.

Instead of answering only from its pre-trained knowledge, the AI first **retrieves relevant information from external sources** and then generates a response.

Simple Definition

RAG means:

Search first → Then Answer

The chatbot first searches the required documents or database and then prepares the answer.

📌 Part 3: Why Do We Need RAG?

Suppose a company has:

- Employee Handbook
- HR Policies
- Product Manuals
- Company Rules
- Internal Documents

ChatGPT does **not automatically know** these private documents.

Instead, we use **RAG**.

The chatbot searches those documents and answers using the retrieved information.

📌 Part 4: Real-Life Example

Imagine a student asks:

What is the fee structure for the AI internship?

The chatbot:

1. Searches the company document.
2. Finds the fee details.
3. Generates an answer based on the document.

Without RAG:

❌ The chatbot may not know the answer.

With RAG:

✅ The chatbot provides the correct answer from the document.

📌 Part 5: RAG Workflow

User Question

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Convert Question into Embedding

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Search Knowledge Base

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Retrieve Relevant Documents

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Send Documents + Question to LLM

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Generate Final Answer

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Return Response to User

📌 Part 6: Components of a RAG System

1. User

Asks a question.

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2. Knowledge Base

Stores:

- PDFs
- Word files
- Excel files

- Text files
- Company documents
- FAQs

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3. Embedding Model

Converts text into numerical vectors so similar meanings can be searched.

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4. Vector Database

Stores document embeddings for fast similarity search.

Examples:

- FAISS
- ChromaDB
- Pinecone
- Weaviate

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5. Large Language Model (LLM)

Reads the retrieved information and generates the final answer.

📌 **Part 7: RAG Architecture**

```
      User
       |
       ▼
  User Question
       |
       ▼
  Embedding Model
       |
       ▼
```

Vector Database

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▼

Retrieve Relevant Documents

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▼

Large Language Model

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▼

Final AI Response

📌 **Part 8: ChatGPT vs RAG**

ChatGPT	**RAG Chatbot**
Uses pre-trained knowledge	Uses external documents
Cannot automatically access company files	Searches company documents
General-purpose	Domain-specific
Limited to training knowledge	Uses latest uploaded information

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📌 **Part 9: Example**

Knowledge Base:

AI Internship Fee: ₹5000

Duration: 3 Months

Location: Technopark

User:

What is the internship fee?

Chatbot searches the document and replies:

The AI Internship fee is ₹5000.

📌 Part 10: Python Example (Concept)

```
documents = [
    "AI Internship Fee is ₹5000",
    "Duration is 3 months",
    "Location is Technopark"
]


question = input("Ask: ")


for doc in documents:
    if "fee" in question.lower() and "Fee" in doc:
        print(doc)
```

Note: This is a simple concept demonstration. Real RAG systems use embeddings and vector databases instead of keyword matching.

📌 Part 11: Real-World Applications

Customer Support

Answer questions from product manuals.

Education

Answer questions from textbooks and study materials.

Healthcare

Retrieve information from medical guidelines.

Banking

Search policy documents and regulations.

Legal

Answer questions from legal documents and contracts.

Company Knowledge Assistant

Search employee handbooks, HR policies, and technical documentation.

📌 Part 12: Advantages of RAG

✅ Uses up-to-date information

✅ Answers from company documents

✅ Reduces hallucinations

✅ No need to retrain the AI model for every document update

✅ Highly scalable

📌 Part 13: Limitations

❌ Depends on document quality

❌ Requires a vector database

❌ Retrieval errors can affect responses

❌ More complex than a basic chatbot

📌 Part 14: Popular RAG Tools

- LangChain
- LlamaIndex
- FAISS
- ChromaDB

- Pinecone
- Weaviate

📌 **Part 15: Theory Questions**

What is RAG?

RAG (Retrieval-Augmented Generation) is an AI technique that retrieves relevant information from external knowledge sources and uses it to generate accurate answers.

Why is RAG used?

To allow AI chatbots to answer questions using custom or private documents instead of relying only on pre-trained knowledge.

What is a Vector Database?

A database that stores text embeddings and enables fast similarity searches.

What is an Embedding?

An embedding is a numerical representation of text that captures its meaning.

Give three applications of RAG.

- Company knowledge assistants
- Customer support systems
- Educational AI tutors

📌 **Part 16: Interview Questions**

What is the difference between ChatGPT and a RAG chatbot?

A standard LLM answers using its training knowledge, while a RAG chatbot retrieves relevant information from external sources before generating an answer.

Why is a Vector Database used in RAG?

It enables efficient retrieval of documents with similar meanings.

What are embeddings?

Embeddings are numerical vectors representing the semantic meaning of text.

Name three vector databases.

- FAISS
- ChromaDB
- Pinecone

Why is RAG important?

It improves answer accuracy and allows AI systems to use the latest company-specific information.

🎯 Tasks for Day 66

🧠 Theory

1. Define RAG.
2. Explain the RAG workflow.
3. Differentiate ChatGPT and RAG.
4. What is an embedding?
5. What is a vector database?

💻 Practical Task 1

Draw the RAG architecture diagram.

💻 Practical Task 2

List five types of documents that can be stored in a knowledge base.

💻 **Practical Task 3**

Research and write short notes on:

- FAISS
- ChromaDB
- Pinecone

💻 **Practical Task 4**

Create a small knowledge base with five questions and answers for a college admission chatbot.

💻 **Practical Task 5**

Explain how a chatbot can answer questions from a PDF using the RAG approach.